

Healthcare Research Assistant Challenge



An AI agent that helps clinical researchers quickly navigate unstructured medical literature and case studies, delivering evidence-based insights and reducing review time from weeks to days.

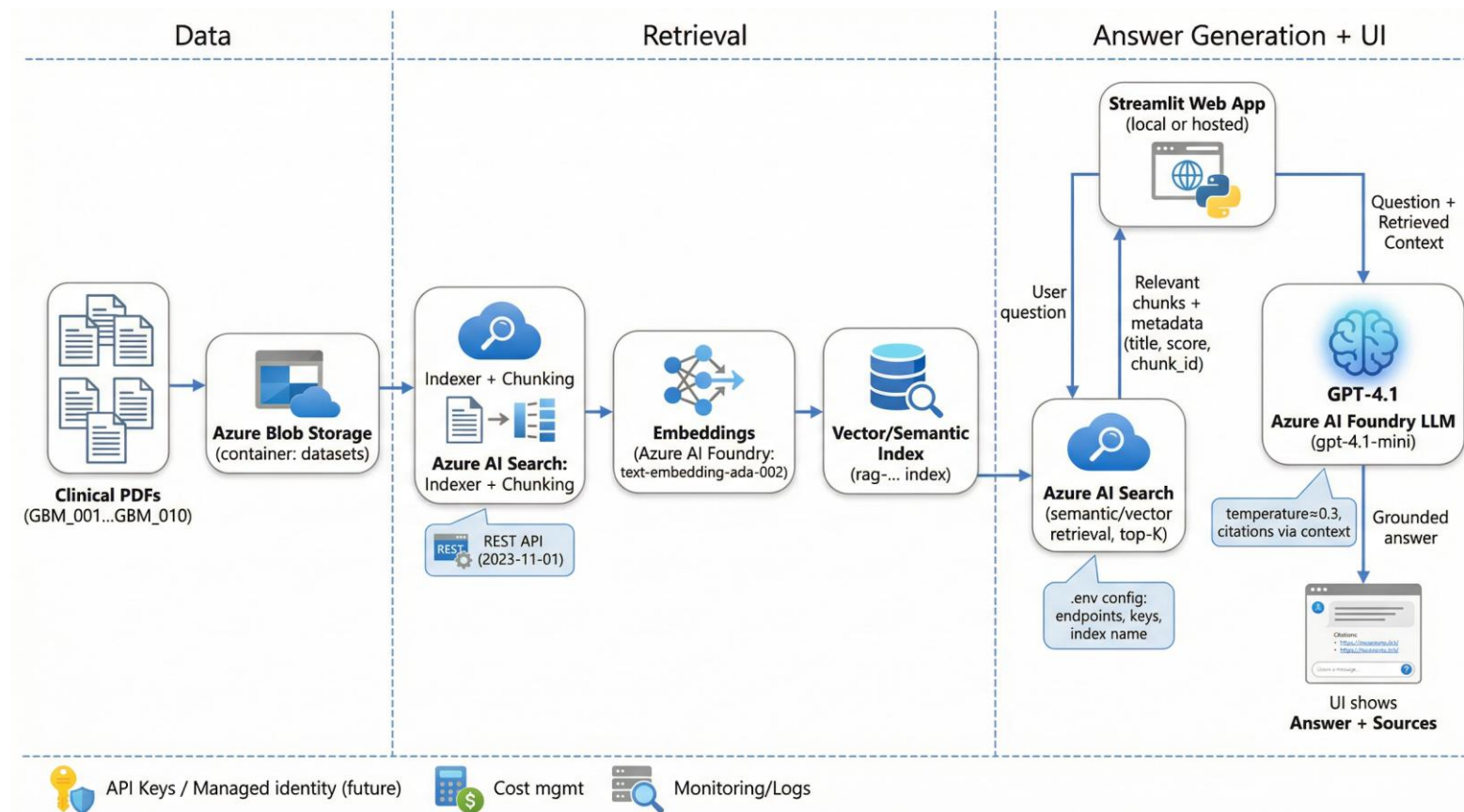
Upload clinical PDFs to Azure Blob Storage (datasets container).

Create Azure AI Search index with chunking + Foundry embeddings.

Web UI app retrieves top-K relevant chunks via Search REST API

Foundry GPT model generates grounded answer using retrieved context

UI shows answer plus sources (title + relevance score)





Future Work / Improvements



FINE-TUNE

ADAPT THE EMBEDDING
MODEL FOR BIOMEDICAL
LANGUAGE AND SECTION-
LEVEL RETRIEVAL



USE A DEDICATED SEARCH STRATEGY

(HYBRID - KEYWORD + VECTOR
+ RERANKING) FOR HIGHER
PRECISION



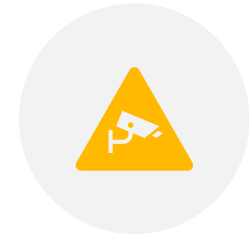
IMPROVE INDEXING

SPLIT PAPERS BY SECTIONS
(ABSTRACT/METHODS/RESULT
S), STORE METADATA, AND
TUNE CHUNK SIZES



ADD GUARDRAILS

TO REDUCE HALLUCINATIONS
(ANSWER-WITH-CITATIONS,
REFUSAL WHEN EVIDENCE IS
MISSING, EVAL TESTS)



PRODUCTION HARDENING

MANAGED IDENTITY,
MONITORING, RATE LIMITS,
AND CACHING